Supporting information

Evaluation of Oral PROTAC Guidelines: Efflux Ratio Outweighs Chameleonicity Descriptors

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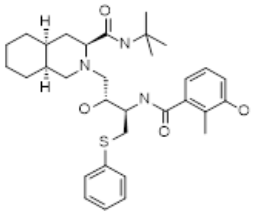
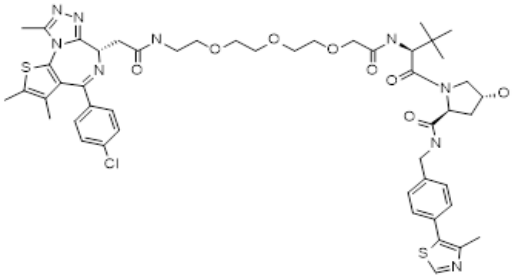
1. Additional figures and tables

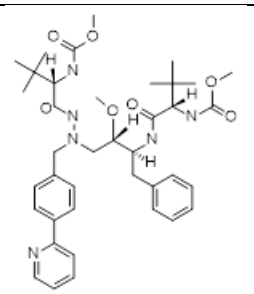
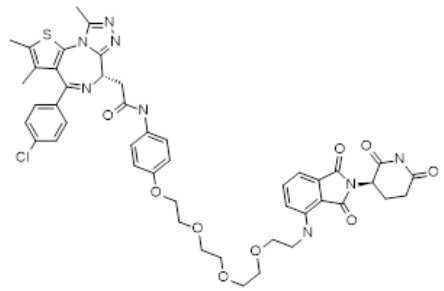
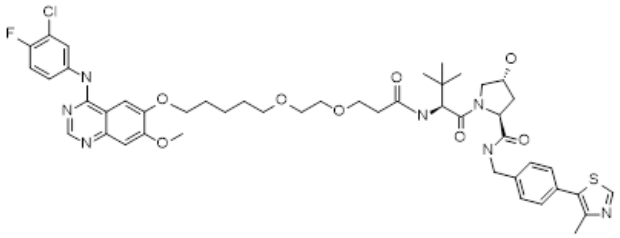
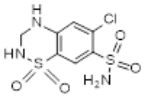
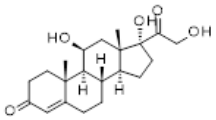
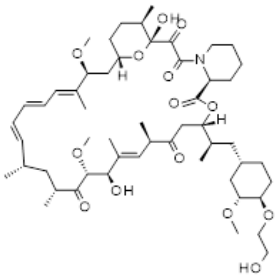
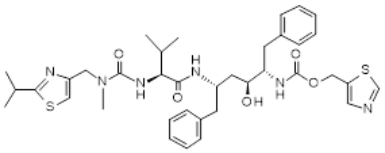
1.1. Table S1 – Mouse oral bioavailability of clinical compounds

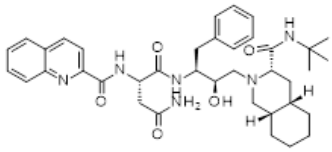
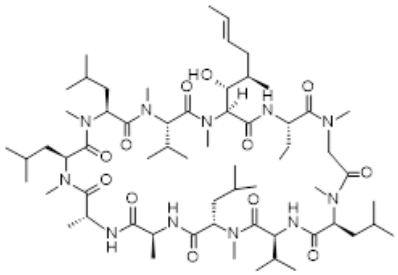
	Literature data			Internal data		
Compound ID	**Mouse F (%)**	**Oral dose (mg/kg)**	**Mouse Strain**	**Mouse F (%)**	**Oral dose (mg/kg)**	**Mouse Strain**
ARV-110[1]	38	5	ICR	65	1	CD1
ARV-471[2]	18	5	ICR	59	1	CD1
ARV-766	Not available			58	1	CD1
KT-474	Not available			24	1	C57BL6
NX-2127[3]	36	10	Not available	45	1	CD1
NX-5948[4]	7-25	10	Not available	74	1	CD1

Note: Differences in oral bioavailability between literature and internal data can be explained by differences in strain, dose, fasted vs fed state, and formulation. It is not uncommon to start seeing saturation of absorption from 10 mg/kg, and it has been shown that fed state usually leads to better oral absorption of protacs than fasted state.

1.2. Table S2. Chamelogk data for reference compounds

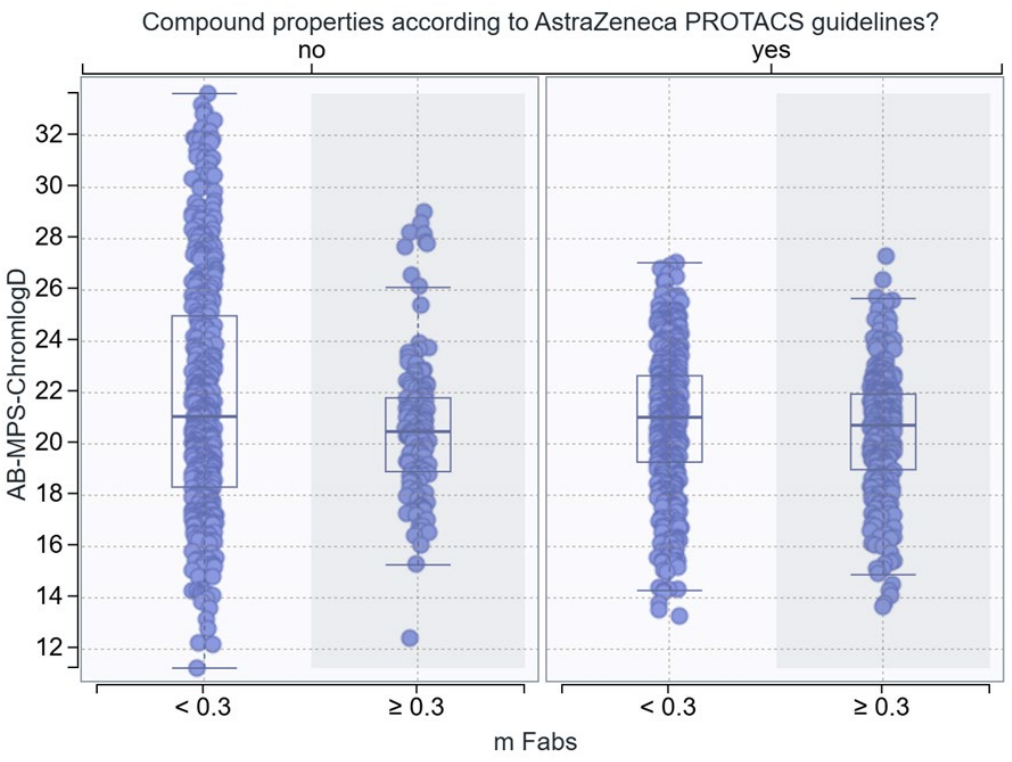
Compound name	Structure	Experimental Chamelogk	Literature Chamelogk
Nelfinavir		0.8	0.74
MZ1		1.5	1.15

Atazanavir		1.0	0.33
ARV-825		1.0	0.72
Gefitinib-Based PROTAC 3		1.3	1.36
Hydrochlothia-zide		0.5	-0.22
Hydrocortisone		0.7	0.100
Everolimus		0.8	0.450
Ritonavir		0.9	0.670

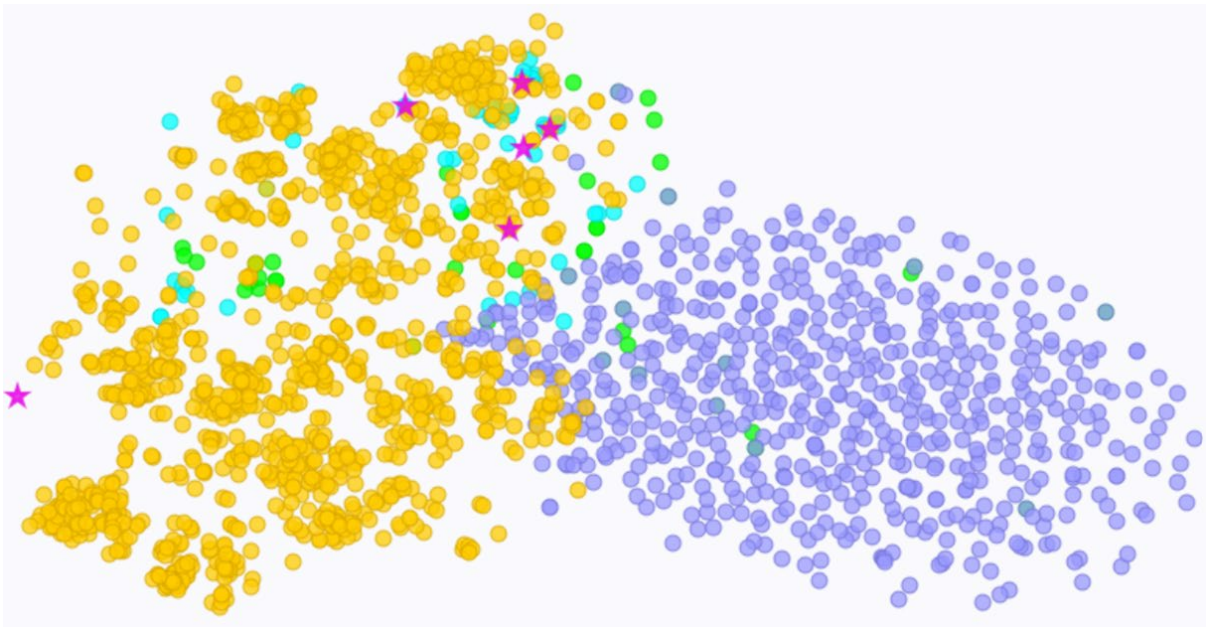
Saquinavir		1.0	1.230
Cyclosporine		1.2	1.250

1.3. Figure S1. Fabs vs CL_u plot within and outside AstraZeneca guidelines

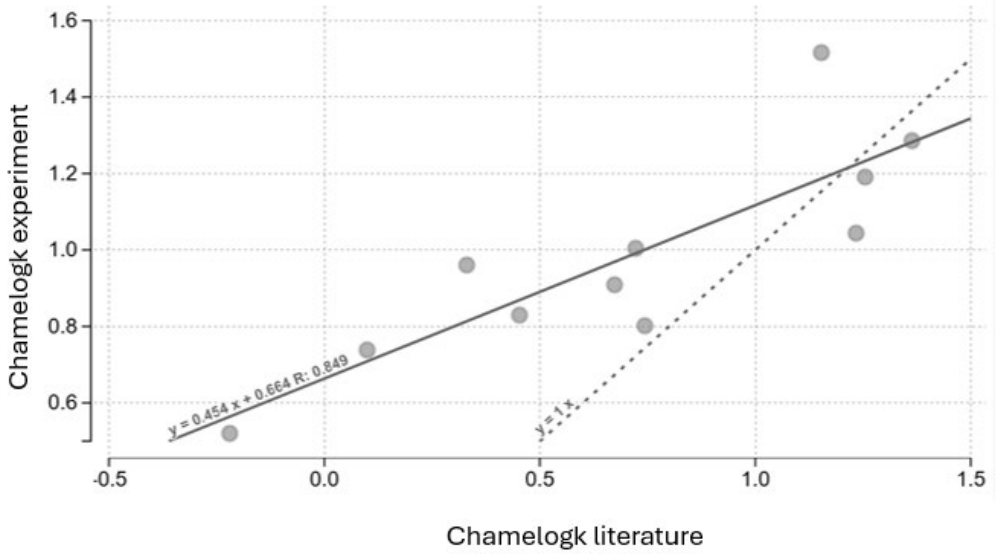
1.4. Figure S2. AB-MPS could not discriminate between poorly and highly absorbed compounds for ~1500 AZ PROTAC dataset.



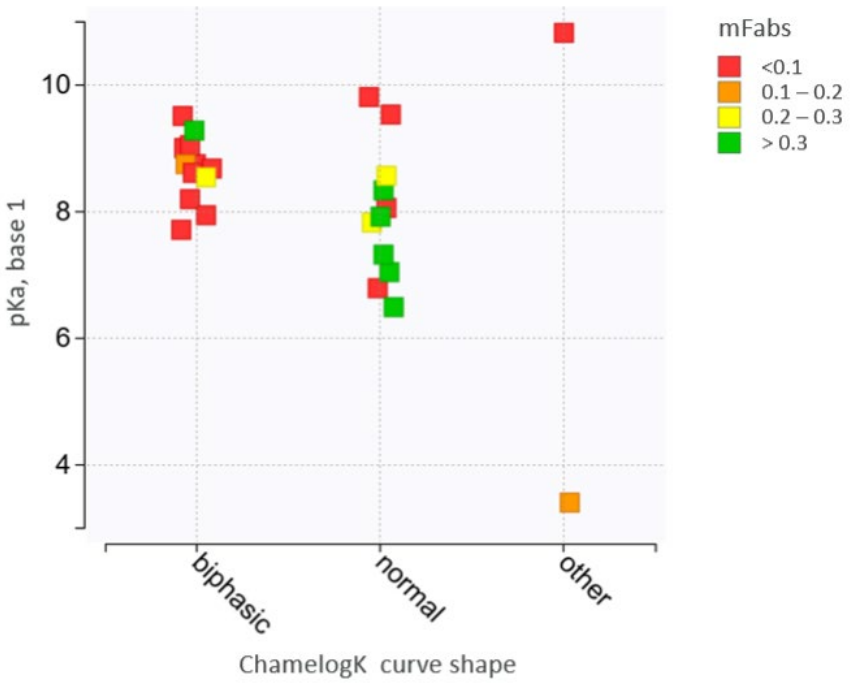
1.5. Figure S3. Similarity map highlighting structural differences between our internal PROTACs dataset and the structures from the publications we have been referring to. Orange: AstraZeneca data set; Pink: Clinical compounds; Purple: disclosed compounds from the AbbVie data set[5]; Green: PROTACs from the Chamelogk data set[6] ; Cyan: disclosed compounds from the BBI data set[7]



1.6. Figure S4. Strong correlation between our experimental values for Chamelogk and the literature values. Based on this the chameleonicity threshold value was adjusted from 0.6 to 0.9



1.7. Figure S5. Chamlelogk curve shape vs pKa



1.8.Molecular dynamics results

Table S3A. Solvent Accessible Surface Area (SASA in Å^2) and 3D Polar Surface Area (3DPSA in Å^2) versus Radius of Gyration (RGYR in Å) for 2 public compounds.

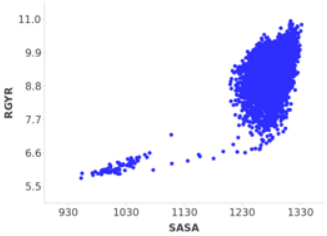
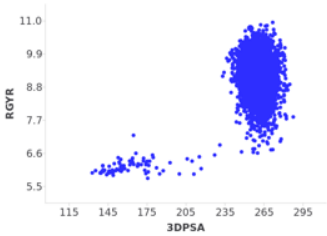
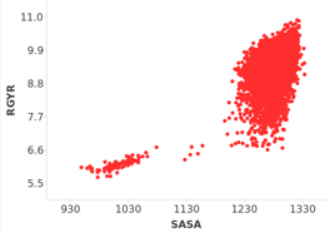
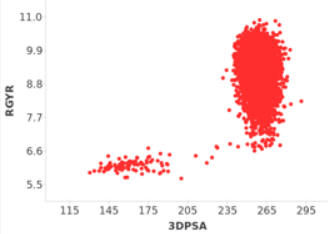
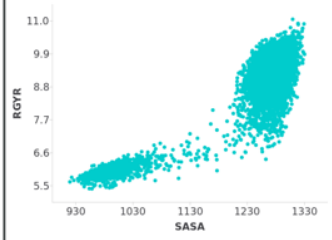
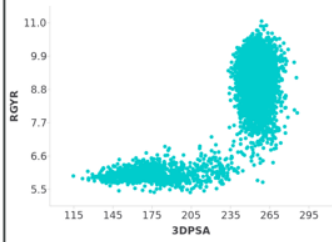
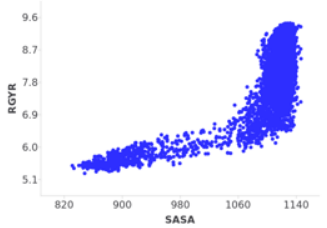
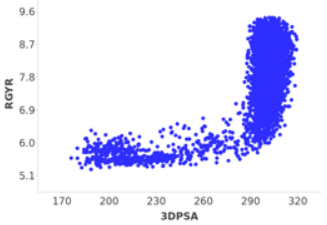
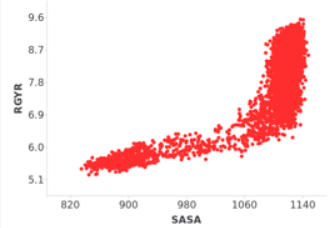
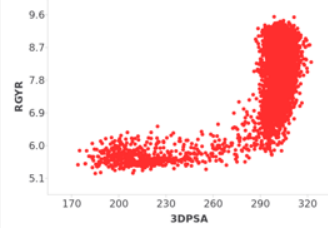
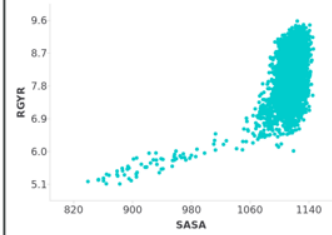
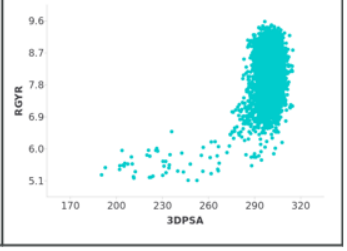
Compound Name	Protonated form in water (Chamelogk < 50%)	Protonated form in acetonitrile (Chamelogk 60-90%)	Not protonated form in acetonitrile (Chamelogk 100%)
KT-474 (normal curve)	 	 	 
NX-2127 (normal curve)	 	 	 

Table S3B. Solvent Accessible Surface Area (SASA in Å2) and 3D Polar Surface Area (3DPSA in Å2) versus Radius of Gyration (RGYR in Å) for first matched pair (compounds 1 and 2) which differs in linker length.

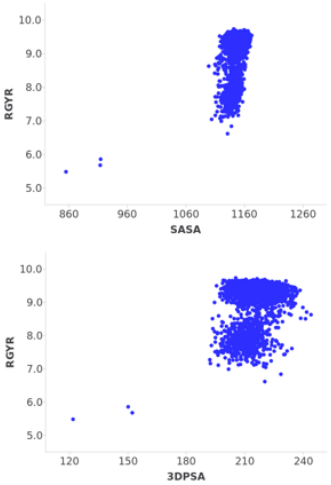
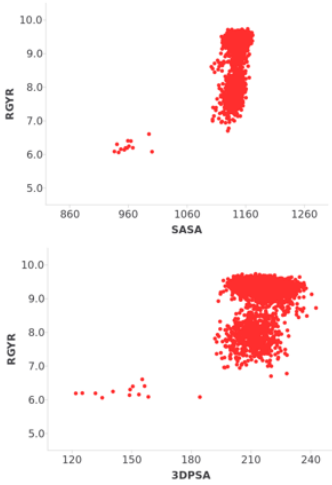
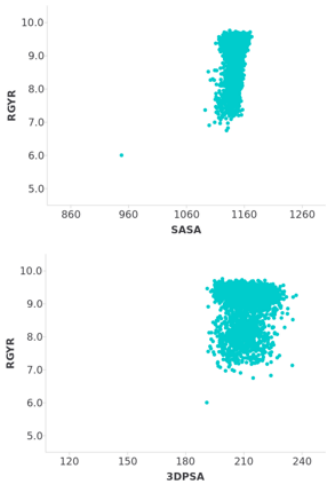
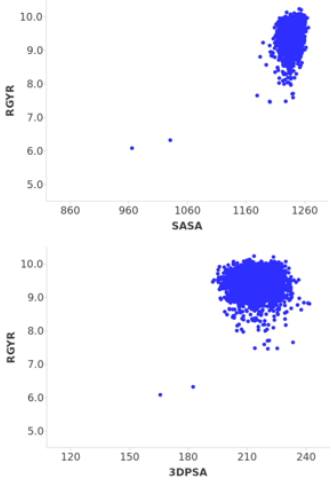
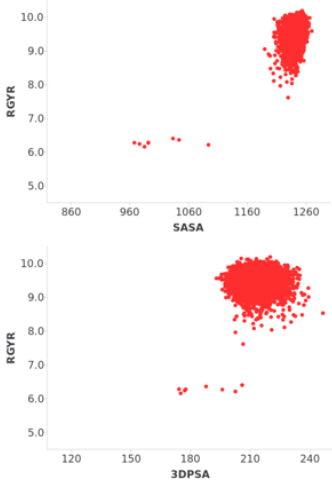
Compound Name	Protonated form in water (Chamelogk < 50%)	Protonated form in acetonitrile (Chamelogk 60-90%)	Not protonated form in acetonitrile (Chamelogk 100%)
1 (normal curve)			
2 (biphasic curve)			

Table S3C. Solvent Accessible Surface Area (SASA in Å2) and 3D Polar Surface Area (3DPSA in Å2) versus Radius of Gyration (RGYR in Å) for second matched pair (compounds 3 and 4) which differs in E3 warhead.

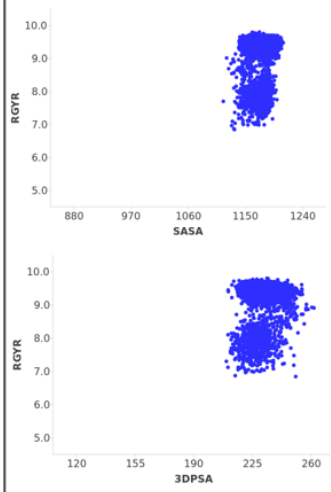
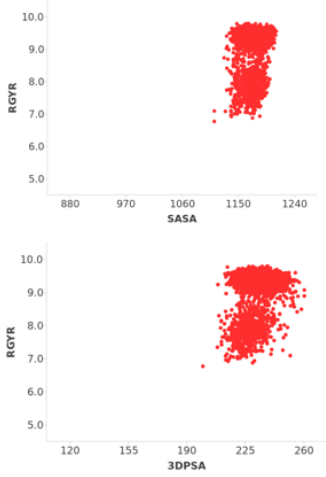
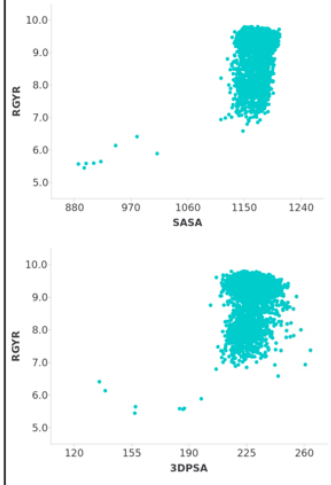
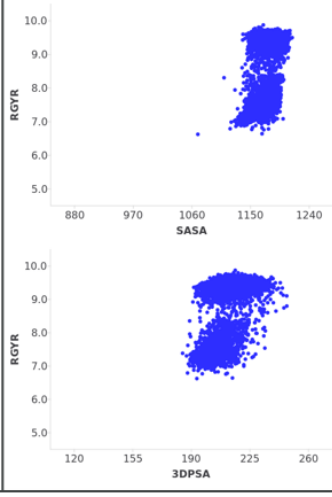
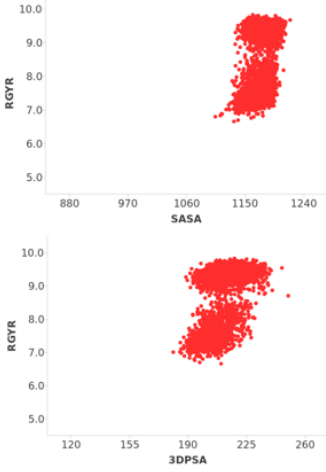
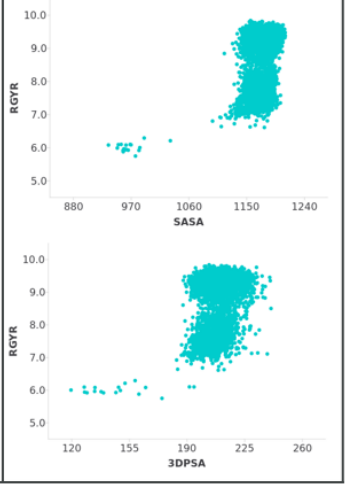
Compound Name	Protonated form in water (Chamelogk < 50%)	Protonated form in acetonitrile (Chamelogk 60-90%)	Not protonated form in acetonitrile (Chamelogk 100%)
3 (normal curve)			
4 (biphasic curve)			

Table S3D. Solvent Accessible Surface Area (SASA in Å2) and 3D Polar Surface Area (3DPSA in Å2) versus Radius of Gyration (RGYR in Å) for third matched pair (compounds 5 and 6) which differs in the vector for E3 warhead.

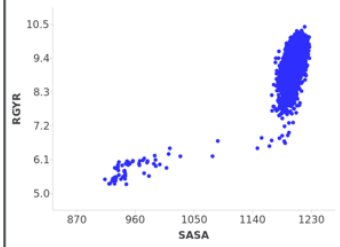
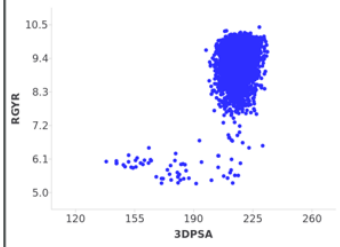
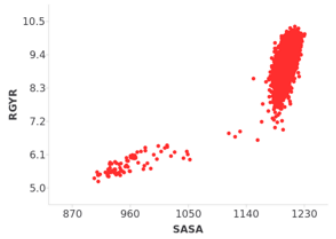
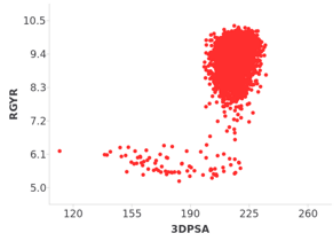
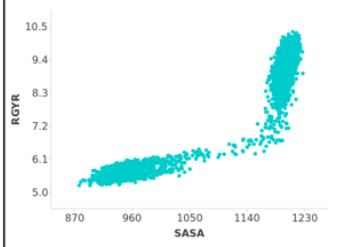
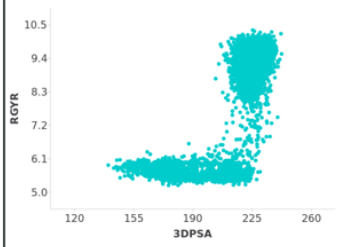
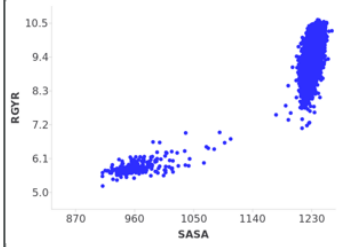
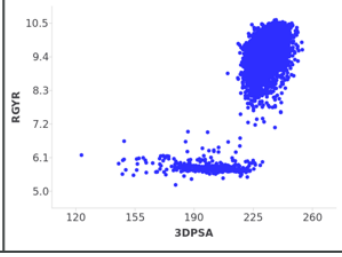
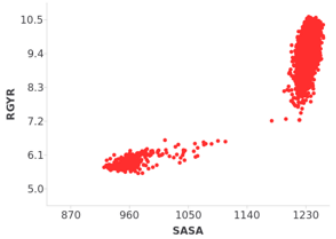
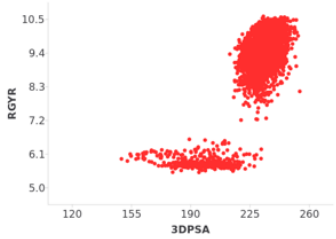
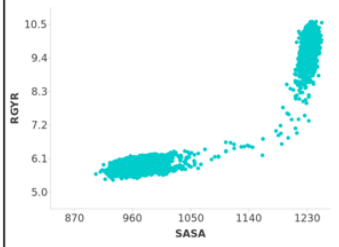
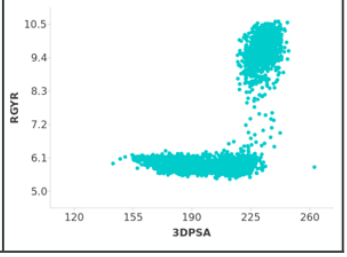
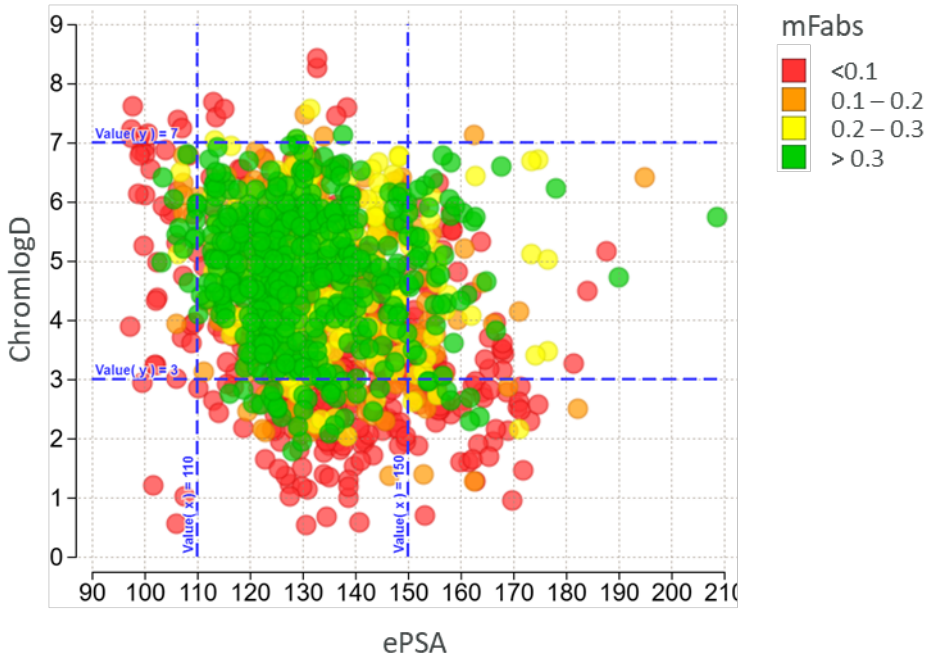
Compound Name	Protonated form in water (Chamelogk < 50%)	Protonated form in acetonitrile (Chamelogk 60-90%)	Not protonated form in acetonitrile (Chamelogk 100%)
5 (normal curve)	 	 	 
6 (biphasic curve)	 	 	 

Table S4. Summary of MD results.

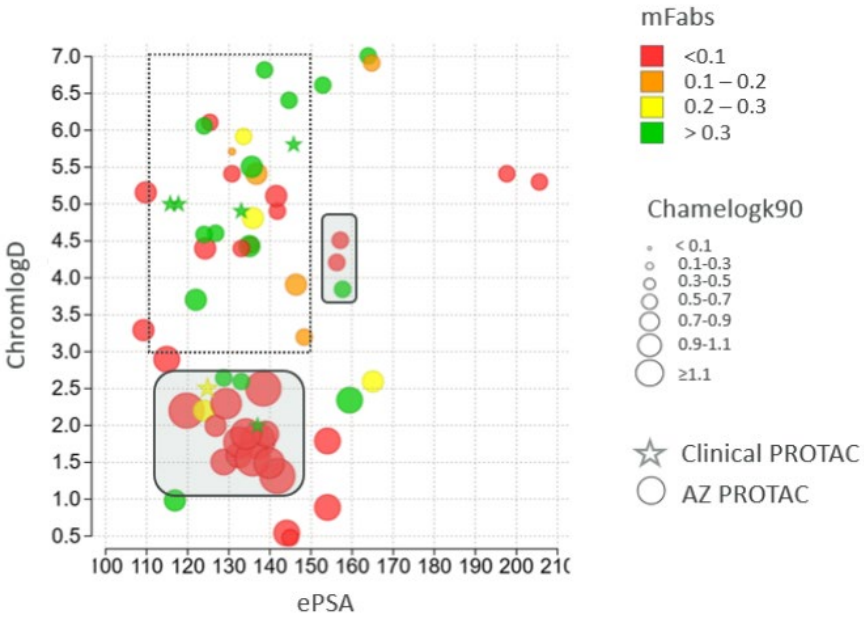
Compound Name	Curve shape	Chamelogk100	Chamelogk90	Comment – what happens going from protonated to not protonated form in acetonitrile	Differences for matched pairs
KT-474	normal	1.70	0.56	More folded conformations	
NX-2127	normal	1.20	0.34	Less folded conformations	
1	normal	2.00	0.60	Folded conformations (singletons) disappear	No difference
2	biphasic	-0.24	0.88	Folded conformations (singletons) disappear	
3	normal	2.00	0.65	Folded conformations (singletons) appear	No difference
4	biphasic	-0.31	0.59	Folded conformations (singletons) appear	
5	normal	1.20	0.40	More folded conformations	No difference
6	biphasic	-0.29	0.60	More folded conformations	

Conclusions:
For analysed matched pairs there is no difference in conformational ensemble when going from protonated to not protonated form in acetonitrile (which mimics going from < 90% ACN to 100% ACN in Chamelogk experiment). Additionally, each matched pair behaves differently when the charge is removed. Therefore, we cannot explain the biphasic curves with conformational analysis.

1.9. Figure S6. Empirical oral space boundaries for internal PROTACs
ChromlogD: 3 – 7; ePSA: 110 – 150

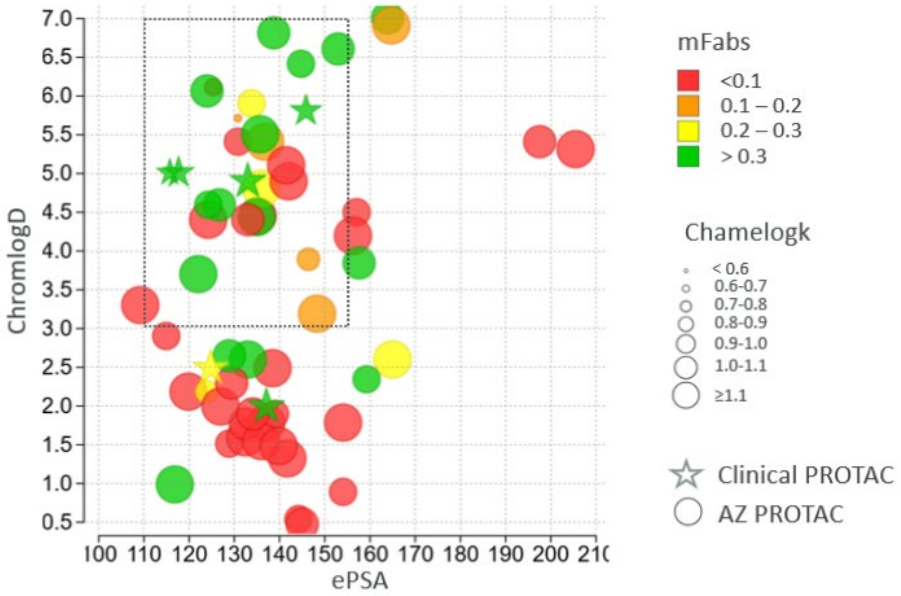


1.10. Figure S7. Chamelogk90 plot



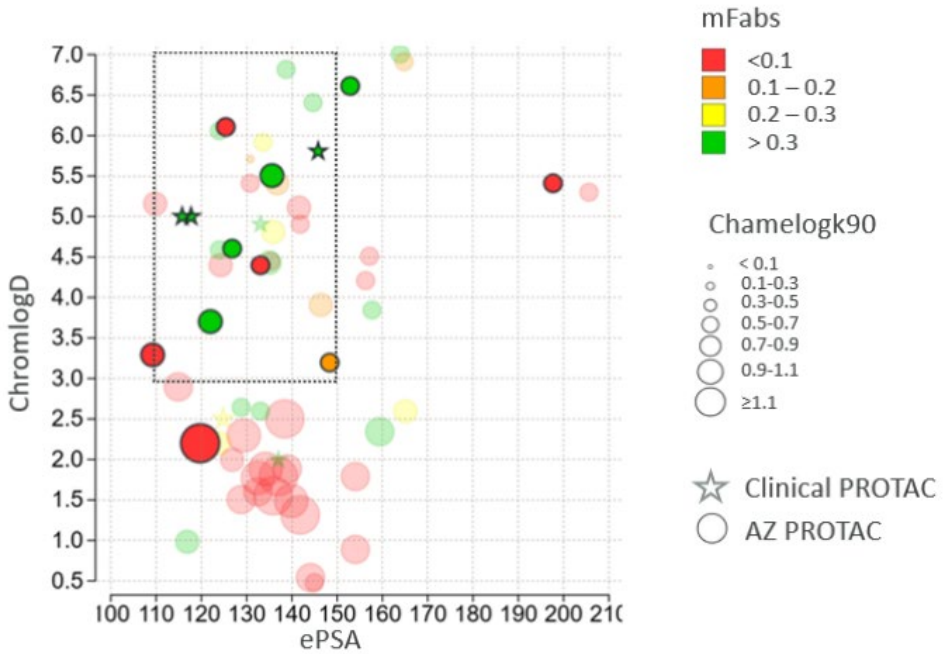
ChromlogD vs ePSA plots, compounds are sized by **Chamelogk90** and coloured by mFabs; The oral zone is highlighted by the dotted rectangle. Increased chameleonicity doesn't rationalize absorption of compounds outside of the oral space – Compounds with the highest Chamelogk90 values are the less absorbed (grey boxes)

1.11. Figure S8. ChamelogkMax plot



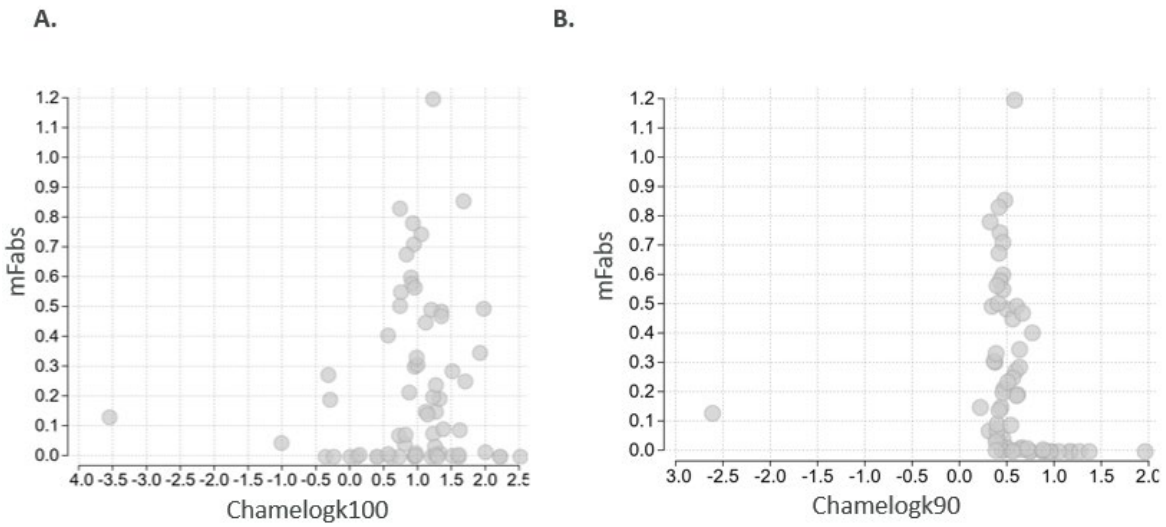
ChromlogD vs ePSA plots, compounds are sized by **ChamelogkMax** and coloured by mFabs; The oral zone is defined by the dotted rectangle. Chameleonicity doesn't rationalize absorption of compounds outside of the oral space – Most compounds have high ChamelogkMax values regardless of whether they are well absorbed or not.

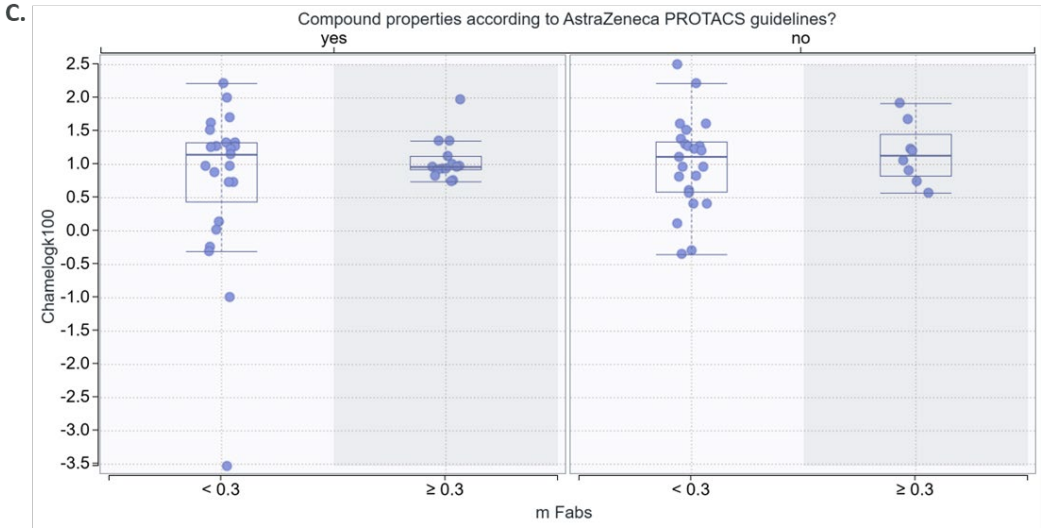
1.12. Figure S9. Neutral compounds, using Chamelogk90

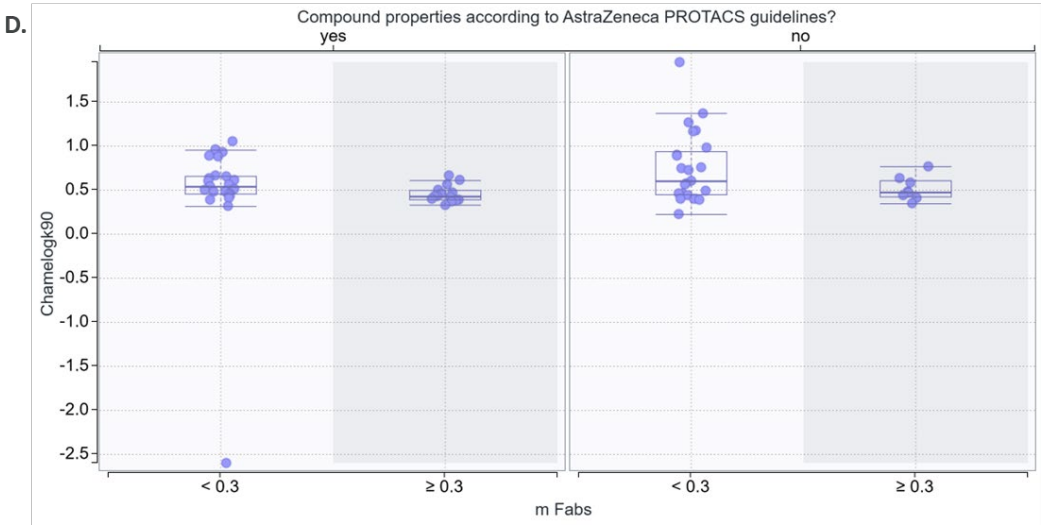


ChromlogD vs ePSA plots, compounds are sized by **Chamelogk90** and coloured by mFabs; Neutral compounds are highlighted; The oral zone is defined by the doted rectangle; Chamelogk90 doesn't rationalize absorption of compounds regardless of ion class.

Figure S10. Correlations plots between Chamelogk and mFAbs



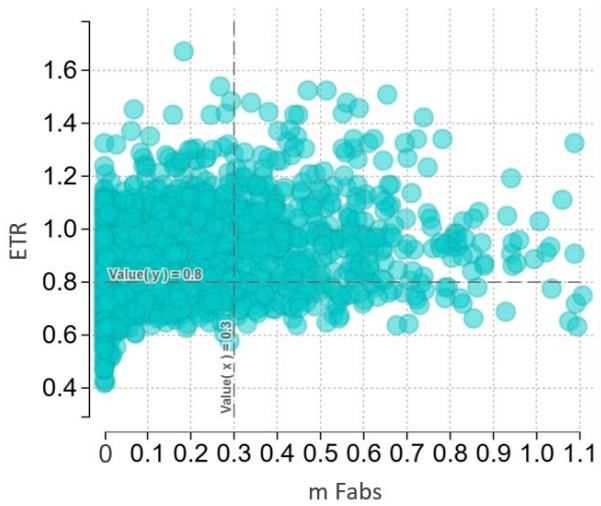




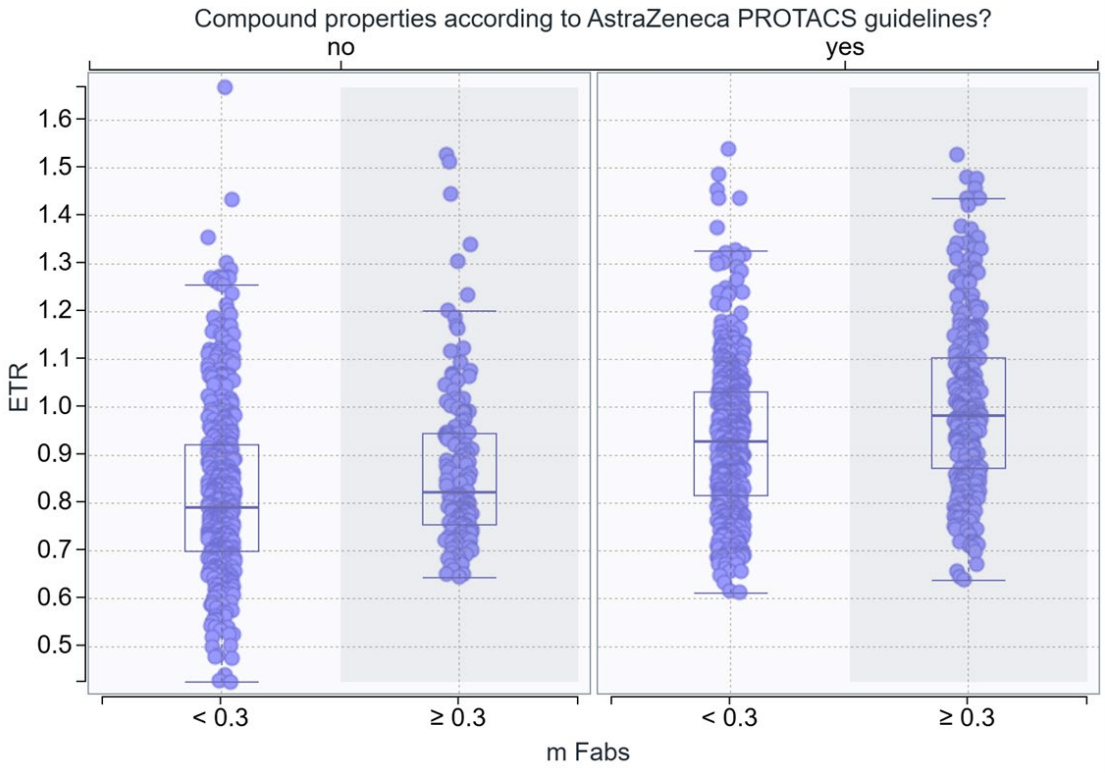
No direct correlation between Chamelogk (100 or 90) and mFabs (**A.** and **B.**), regardless of whether compounds adhere to AstraZeneca guidelines for oral PROTACs or not (**C.** and **D.**)

Figure S11:

A. With our PROTAC dataset, 77 % of the highly absorbed compounds (m Fabs > 0.3) have an ETR > 0.8

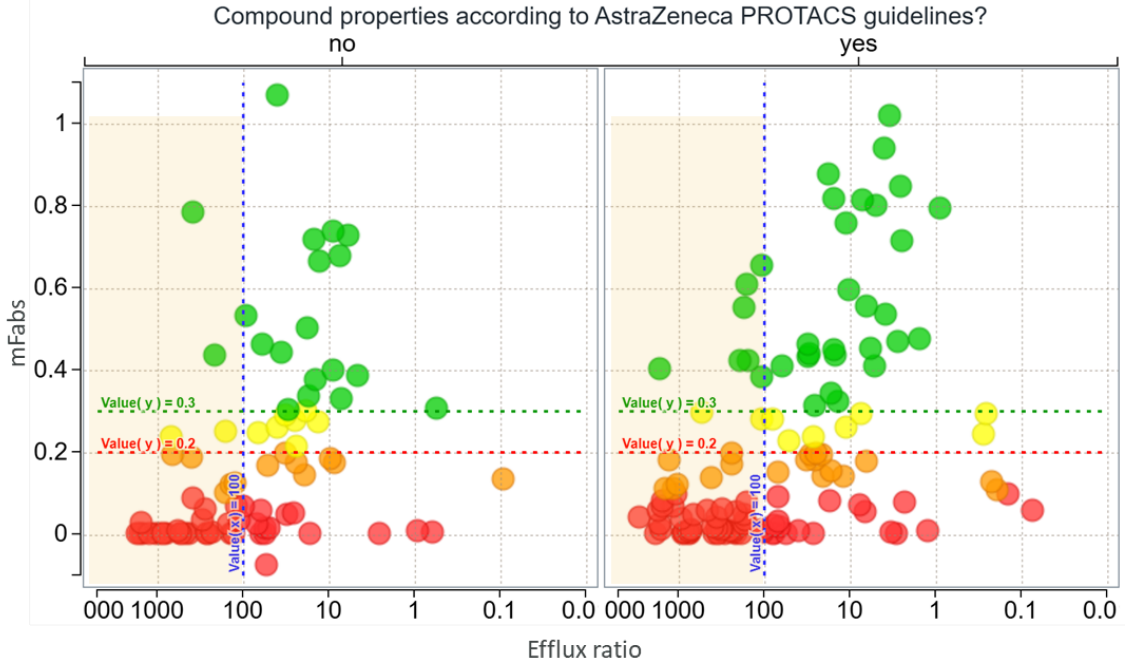


B. ETR didn't help to refine our internal guidelines



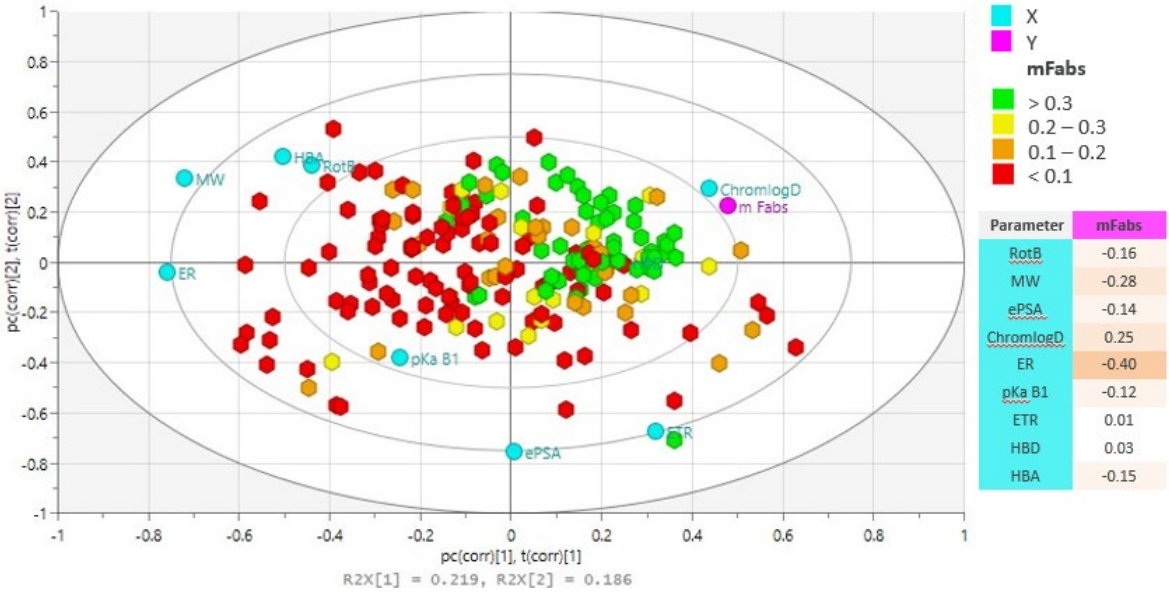
1.13. Figure S12:

A. Efflux ratio vs mFabs plot outside or inside AstraZeneca guidelines. 89% (resp. 83%) of the compounds with ER > 100 have a mFabs < 20% outside (resp. inside) the AstraZeneca guidelines.



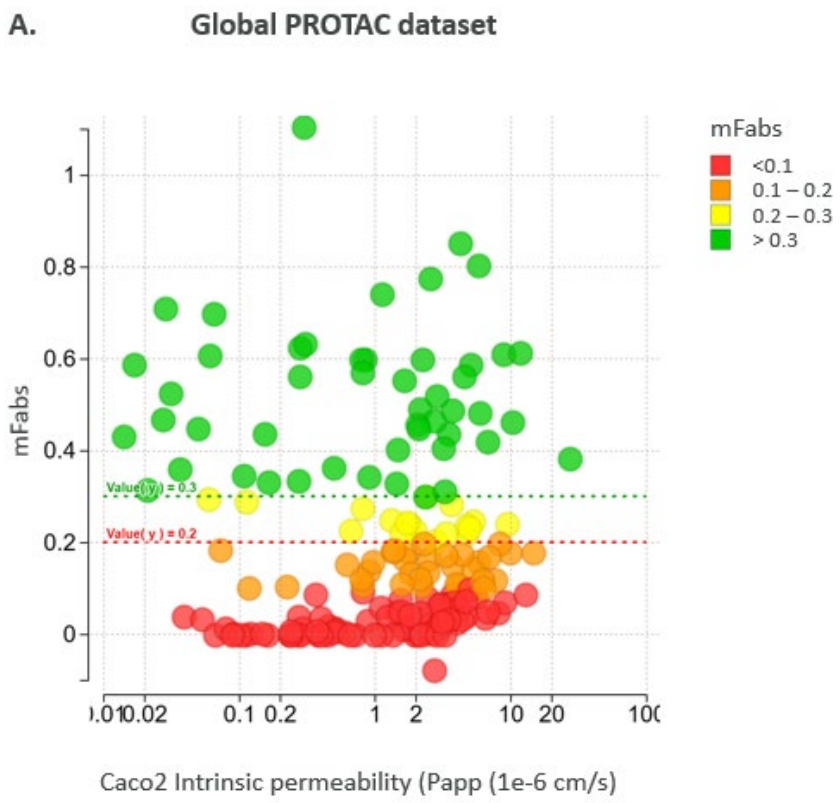
B. Same dataset as above, Efflux ratio vs mFabs

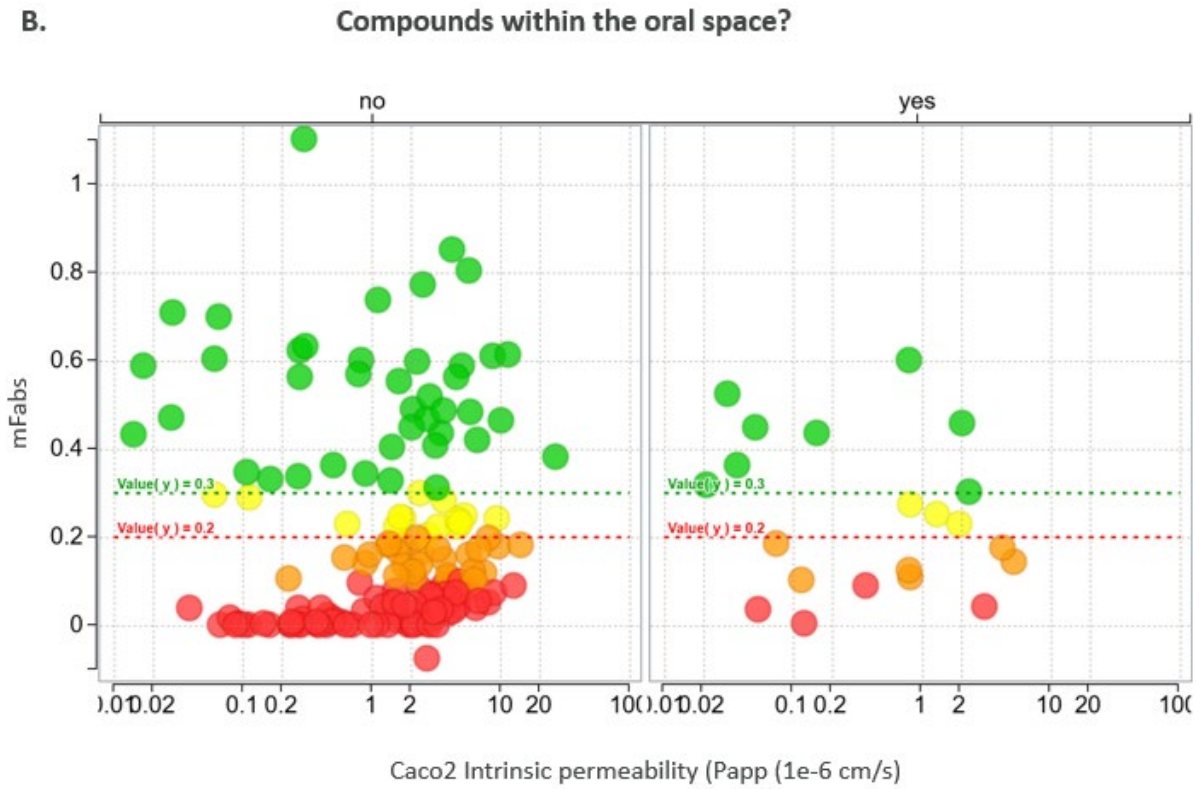
Figure S13: Partial Least Square (PLS) biplot highlighting relationships between the fraction absorbed of each compound, properties and mFabs. ER is strongly correlated to mFabs as shown in the correlation matrix.



Parameter	mFabs
RotB	-0.16
MW	-0.28
ePSA	-0.14
ChromlogD	0.25
ER	-0.40
pKa B1	-0.12
ETR	0.01
HBD	0.03
HBA	-0.15

1.14. Figure S14: mFabs vs Intrinsic permeability plots





2. Experimental methods

2.1. Chamelogk

Chamelogk measurements were performed at o2h, India (https://o2h.com) using the method described by Garcia Jimenez et al.[6]
We also calculated the Chamelogk90 from the raw data as described below.

Briefly, Log k' was measured with 6 different mobile phases for each compound using a PLRP-S column.

Test Compound	Time (min)	Isocratic condition	A% 20 mM Ammonium Acetate Buffer	B% Acetonitrile
All compounds	15	1	50	50
		2	40	60
		3	30	70
		4	20	80
		5	10	90
		6	0	100

The extrapolated values at 90 and 100% ACN (Ext. log k' 90, respectively 100) were calculated as below, where the slope and intercept were obtained from the linear regression plot with experimental values at 50, 60, and 70% ACN.

Log k' (at 90 % acetonitrile) = (Slope x 90) + Intercept
Log k' (at 100 % acetonitrile) = (Slope x 100) + Intercept

Chamelogk90 and 100 were calculated from the experimental and extrapolated values as below:

Chamelogk90 = Exp. log k' 90 - Ext. log k' 90
Chamelogk100 = Exp. log k' 100 - Ext. log k' 100

The results and curves for the test compounds are compiled in the associated excel file.

2.2. In vitro DMPK assays

ePSA and **Chromlog D:** The data was generated as previously reported[8].
Intrinsic Caco2 Papp and Caco2 efflux[9]
An automated assay to study Caco-2 monolayer permeability in the apical to basolateral direction including a pH gradient (pH 6.5 in apical donor compartment and pH 7.4 in basolateral receiver compartment) was used, with addition of a transport inhibitor cocktail to determine the passive (intrinsic) permeability across the membrane. The inhibitor cocktail contained 10 µM zosuquidar, 30 µM benzbromarone and 2 µM KO-143.
For efflux determination, a similar set-up with no pH gradient (pH 7.4 in both apical and basolateral compartment) and no inhibitor cocktail was used. Efflux ratios were calculated as the ratio of the apparent permeability coefficient in the apical to basolateral direction and the apparent permeability coefficient in the basolateral to apical direction: ER = PappB-A/PappA-B.

High throughput FeSSIF solubility was determined using the same protocol as previously described for high throughput aqueous solubility, starting from a DMSO stock solution, where DMSO is evaporated[10], replacing the PBS buffer with FeSSIF media**.**
pKa (compounds in figure S1) was measured at Pion (https://www.pion-inc.com/solutions/analytical-services)

2.3. Predicted properties

tPSA was calculated by the Ertl method using the software implementation from Openeye
There are no sources in the current document.
Scientific, Cadence Inc.35
pKa model was built using internal data on 25K compounds, mostly with low MW and 1-2 ionic centres.
Rotatable bound count was calculated using OESelma. Rotatable bound is defined as any single non-ring bound, bounded to nonterminal heavy (i.e., non-hydrogen) atom. It considers the carbon-carbon triple bound in acetylene rotatable. The function excludes single bounds in certain groups suh as sulfonamides,esters and amidine.
HBD and **HBA** were calculated using OESelma.
Machine learning models: Machine learning models for ePSA and ChromLogD were trained and evaluated as previously reported[8].

2.4. Similarity map

The similarity map (Figure S3) was generated in D360 (25.5) (Certara Scientific Informatics Platform) by calculating the chemical similarity between compounds and visualizing these relationships. The platform leverages RDKit for cheminformatics, allowing for the generation of Tanimoto similarities on fingerprints like Morgan (ECFP4) to determine structural closeness. To visualize high-dimensional similarity data, D360 uses Dimensionality Reduction techniques like Principle Component Analysis (PCA) to map compounds onto a 2D scatter plot, where proximity indicates high similarity.
The axes are usually unlabelled in terms of specific physicochemical properties, but the proximity of points indicates higher structural similarity. In these maps, spatial closeness corresponds to similarity in structure, whereas large distances indicate dissimilarity.

2.5. In vivo mouse pharmacokinetics

Mouse PK parameters were obtained from non-compartmental analysis (NCA) after IV and PO dosing in C57 BL/6 or CD1 mice at Pharmaron, China. The animals were housed in an Association for Assessment and Accreditation of Laboratory Animal Care (AAALAC) accredited animal (accreditation number: 001322) facility in a controlled environment (20–25 °C and 40–70% relative humidity) with a 12 h light/12 h dark cycle with free access to food and water. Pharmaron's (Beijing Co., Ltd., China) Institutional Animal Care and Use Committee (IACUC) approved the studies, and the animal experiments conform to all relevant regulatory standards. Compounds were dosed intravenously (0.5 mg/kg) and orally (1 mg/kg) in cassettes of five compounds using a standard formulation made of 5% DMSO and 95% sulfobutylether-β-cyclodextrin (SBE-B-CD) (30% w/v) in water, with the pH adjusted to around 4. This formulation was chosen as a potentially clinically relevant enabling formulation. Blood samples were collected from the dorsal metatarsal vein at different time points after administration (0.03, 0.08, 0.17, 0.5, 1, 2, 4, 8, and 24 h for IV and 0.08, 0.25, 0.5, 1, 2, 4, 8, and 24 h) and treated with EDTA-K2 to avoid coagulation, centrifuged at 4,000g for 5 min at 4 °C to obtain plasma that was stored at −75 ± 15 °C prior to analysis. After protein

precipitation, the plasma concentration was determined by LC/MS/MS and PK parameters were calculated using NCA data analysis.

2.6. PLS analysis:

Partial Least Squares (PLS) analysis was performed with 201 PROTACs using ePSA, ChromlogD, HBD, HBA, MW, number of rotationable bounds, pKa base1, efflux ratio and ETR (ePSA/TPSA ratio) as X parameters and mFabs as Y parameter. PLS biplots and the associated correlation matrix were generated in SIMCA17 software (https://www.sartorius.com/).

2.7. Molecular dynamics

REST-MD (Replica Exchange with Solute Tempering) calculations in explicit solvent were done using Schrodinger Suite (Schrödinger Release 2024-4: Maestro, Schrödinger, LLC, New York, NY, 2024, Schrödinger Release 2024-4: Desmond Molecular Dynamics System, D. E. Shaw Research, New York, NY, 2024. Maestro-Desmond Interoperability Tools, Schrödinger, New York, NY, 2024) with OPLS4 (Optimized Potentials for Liquid Simulations family) force field. For torsional angles not explicitly included in OPLS4, we derived parameters from quantum mechanical calculations and incorporated them into the OPLS4 framework. For protonated PROTAC species, chloride counterions were added to achieve overall charge neutrality. Moreover, all simulations were conducted with 0.15M NaCl to mitigate artefacts associated with periodic boundary conditions.

To mimic the Chamelogk experiment we performed 3 separate runs for each compound: using (1) protonated form in water, (2) protonated form in acetonitrile (which corresponds to a situation when we still have water in the system, that can protonate the molecule, but the majority of the solvent is acetonitrile), and (3) not protonated form in acetonitrile (which corresponds to 100% of aprotic solvent).
Since we observe biphasic behaviour only for basic compounds, we have chosen PROTACs with a base in the linker for this analysis. We have selected 3 different matched pairs from our AZ collection, one with a normal curve and the other with biphasic curve. Additionally, we have run the same simulations for 2 public compounds (that are basic but display normal curves) to see possible similarities and differences.
To analyse the flexibility of investigated PROTACs we plotted Solvent Accessible Surface Area (SASA) and 3D Polar Surface Area (3DPSA) versus Radius of Gyration (RGYR) for all conformers from each run. All the plots are collected in Table S3 and the results are summarised in Table S4.

References

(1) Nguyen, T. T. L.; Kim, J. W.; Choi, H. I.; Maeng, H. J.; Koo, T. S. Development of an LC-MS/MS Method for ARV-110, a PROTAC Molecule, and Applications to Pharmacokinetic Studies. *Molecules* **2022**, *27*, 1977. https://doi.org/10.3390/molecules27061977.

(2) Choi, H. I.; Choi, J.; Kim, J. W.; Lee, Y. H.; Cho, K. H.; Koo, T. S. Stability Evaluation and Pharmacokinetic Profiling of Vepdegestrant in Rodents Using Liquid Chromatography–Tandem Mass Spectrometry. *Molecules* **2024**, *29*, 4048. https://doi.org/10.3390/molecules29174048.

(3) Jeffrey Mihalic; Timothy Ingallinera; Daisuke Kato; Hao Lu; Jun Ma; Joel McIntosh; Mark A. Noviski; Ge Peng; Luz Perez; Ryan Rountree; May Tan; Austin Tenn-

McClellan; Jeffrey Wu; Jordan Ye; Stephanie Yung. *First Disclosure of NX-2127, an Oral Targeted of Bruton's Tyrosine Kinase (BTK) with concurrent Immunomodulatory Activity for the treatment of B-Cell Malignancies*; 2023. https://ir.nurixtx.com/static-files/84b651a2-ae3e-4b4f-93c8-28a0d03972ff (accessed 2025-11-18).

(4) Mihalic, J. *First Disclosure of NX-5948, an Oral Targeted Degrader of Bruton's Tyrosine Kinase (BTK) for the Treatment of B-Cell Malignancies*; 2023. https://www.nurixtx.com/wp-content/uploads/2023/08/NX5948-First.Disclosure-ACS-2023-Final.pdf (accessed 2025-11-18).

(5) Price, E.; Weinheimer, M.; Rivkin, A.; Jenkins, G.; Nijsen, M.; Cox, P. B.; DeGoey, D. Beyond Rule of Five and PROTACs in Modern Drug Discovery: Polarity Reducers, Chameleonicity, and the Evolving Physicochemical Landscape. *J. Med. Chem.* **2024**, *67*, 5683–5698. https://doi.org/10.1021/acs.jmedchem.3c02332.

(6) Garcia Jimenez, D.; Vallaro, M.; Rossi Sebastiano, M.; Apprato, G.; D'Agostini, G.; Rossetti, P.; Ermondi, G.; Caron, G. Chamelogk: A Chromatographic Chameleonicity Quantifier to Design Orally Bioavailable Beyond-Rule-of-5 Drugs. *J. Med. Chem.* **2023**, *66*, 10681–10693. https://doi.org/10.1021/acs.jmedchem.3c00823.

(7) Baylon, J. L.; Chalkley, M. J.; Siu, T.; Shou, W.; Sun, Y.; Cai, X.; Paiva, A.; Patel, S.; Zvyaga, T.; Weiss, D. R. New Multiparameter Index Is a Strong Predictor of Oral Bioavailability for Heterobifunctional Degraders. *ACS Med. Chem. Lett.* **2025**, *16*, 1108–1113. https://doi.org/10.1021/acsmedchemlett.5c00156.

(8) Santisteban Valencia, Z.; Kingston, J.; Miljković, F.; Rowbottom, H.; Mann, N.; Davies, S.; Ekblad, M.; Di Castro, S.; Kwapień, K.; Malmerberg, E.; Friis, S. D.; Lundbäck, T.; Leek, T.; Wernevik, J. Closing the Design-Make-Test-Analyze Loop: Interplay between Experiments and Predictions Drives PROTACs Bioavailability. *J. Med. Chem.* **2024**, *67*, 20242–20257. https://doi.org/10.1021/acs.jmedchem.4c01642.

(9) Fredlund, L.; Winiwarter, S.; Hilgendorf, C. In Vitro Intrinsic Permeability: A Transporter-Independent Measure of Caco-2 Cell Permeability in Drug Design and Development. *Mol. Pharm.* **2017**, *14*, 1601–1609. https://doi.org/10.1021/acs.molpharmaceut.6b01059.

(10) Wernevik, J.; Bergström, F.; Novén, A.; Hulthe, J.; Fredlund, L.; Addison, D.; Holmgren, J.; Strömstedt, P. E.; Rehnström, E.; Lundböck, T. A Fully Integrated Assay Panel for Early Drug Metabolism and Pharmacokinetics Profiling. *Assay Drug Dev. Technol.* **2020**, *18*, 157–179. https://doi.org/10.1089/adt.2020.970.